

ProtoDUNE HD + VD — Wire-Cell Noise Filtering

one OmnibusNoiseFilter cascade, two filter sets: per-channel baseline → detector extras → coherent common-mode subtraction



	resampler applies to	per-channel stage	detector-specific stage	coherent groups	implementation
PD-HD	all 4 APAs (512 ns DAQ)	PDHDOneChannelNoise	FEMBNoiseSub — coherent negative-pulse dips (before ③)	per FEMB: 40 ch U/V · 48 ch W	sigproc ProtoduneHD.cxx
PD-VD	bottom CRPs only (top: tick relabel 500 ns)	PDVDOneChannelNoise + RMS ∉ [1,60] → noisy	ShieldCouplingSub — top-CRP U shield strips (after ③)	per conduit: 16–48 ch U/V FR⊗E deconv protection	sigproc ProtoduneVD.cxx

PD-HD: `cfg/.../pdhd/{nf,chndb-base}.jsonnet` · see `pdhd/docs/nf-chain-diagram.md`

PD-VD: `cfg/.../protodunevd/{nf,chndb-base,chndb-resp-*}.jsonnet` · see `pdvd/docs/20_pdvd-nf-chain-diagram.md`